Use of virtual laboratories

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INDEX

Use of virtual laboratories **1**
INDEX **2**
1. Introduction **3**
2. Registration in the system **4**
3. System access **5**
3.1 Access to the environment 5
3.2 Access to equipment 6
3.3 Using remote equipment 7
3.4 Saving data 9
3.5 Possible errors 9
3.6 Log out 9
4. Recommended Applications **11**
4.1 Web Browsing 11
4.2 Programming Environments 11
4.2 Terminal 12
4.3 Python notebook: 12

1. Introduction

The Department makes available to students a series of resources that can be accessed remotely to carry out activities and subjects without requiring their physical presence in the classrooms.

The only access condition required is to have a valid University account., with any profile (student, teacher...). There are two types of resources:

- Physical resources, that is, the student will connect remotely to a physical computer. In this type of resource you will have the possibility of using a graphics accelerator GPU in Linux.
- Virtual resources, that is, virtual machines with Windows that will allow access to programs that require a license.

In this manual, only connecting to resources of the first type will be covered, but the preliminary steps (registration and access) will be similar.

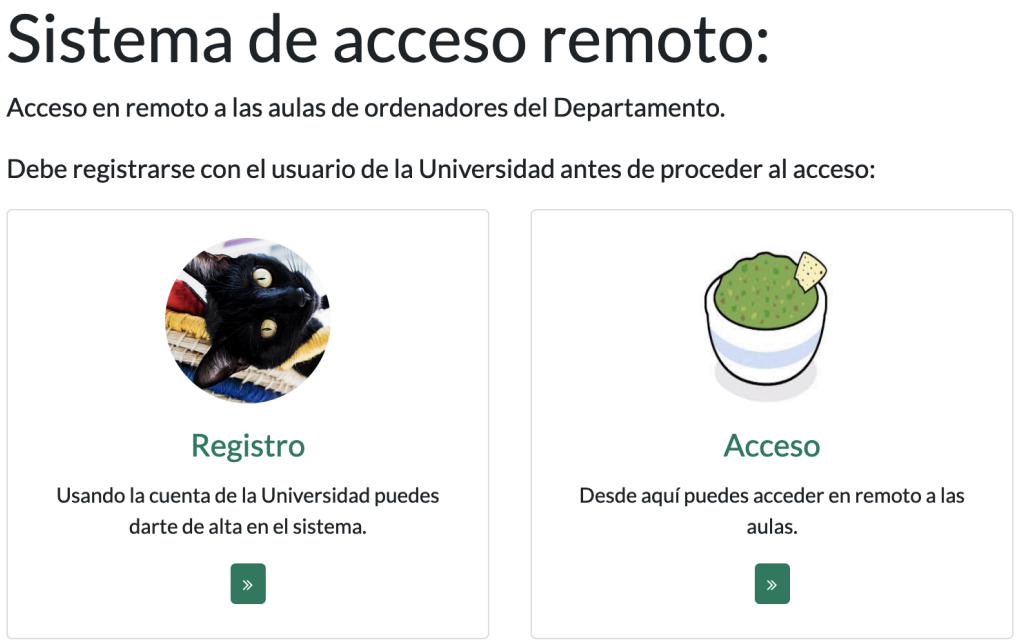
- In order to use remote resources we must complete two simple steps:
- Register for the service with our University account. Doing so will create an account in the system, with the indicated password.

Access the resources with the account created in the previous point.

In the following points we will show in detail how to proceed.

2. Registration in the system

To register in the system we must go to the following address, https://pixel.tsc.uc3m.es/ where the following screen will be shown to us:

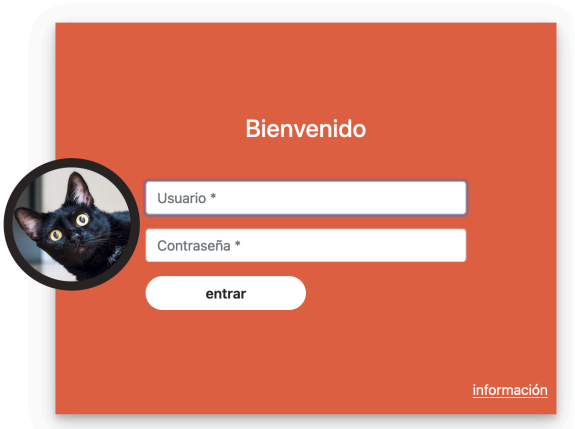


As you can see, it offers the two options mentioned above, on the one hand, register (the first step), and on the other, log in (the second step).

If we click on the registration sale, we begin the process. Remember that you will need a valid University account to complete the process.

If we follow the steps, the system will ask us to accept the conditions and generate a new password, after which we can move on to the next step, accessing the system.

As can be seen in the image below, if the user was already registered in the system, the password will be updated.

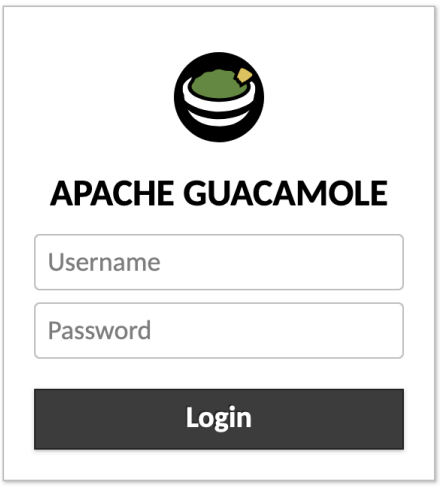


3. System access

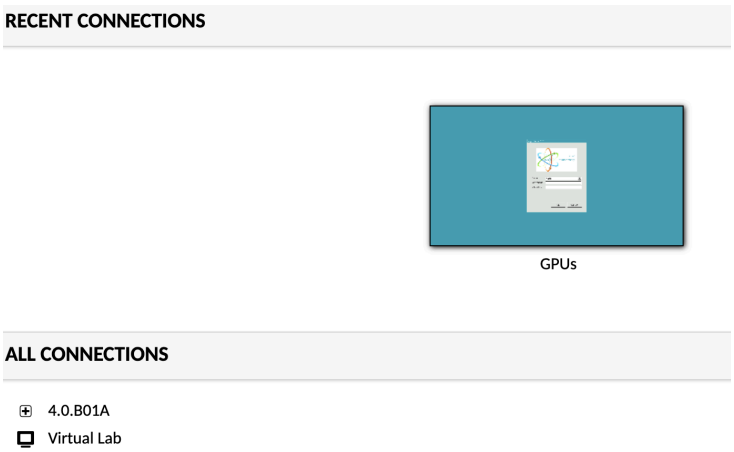
3.1 Access to the environment

Once we have correctly registered in the system, we can use the university username and password to access the remote laboratories.

To do this, we return to the main page, https://pixel.tsc.uc3m.es/ and we choose the second option, “access”, which will present us with the following screen:



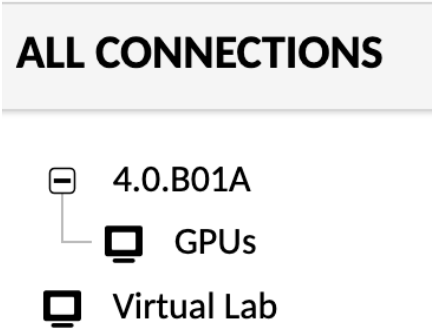
Once we have entered the remote laboratories, it will present us with a screen in which we can see the recent connections, if we have them, and the available remote laboratories:



We see that we have two different groups:

- Physical resources, that is, the student will connect remotely to a physical computer. In this type of resource you will have the possibility of using a graphics accelerator GPU in Linux. **It is the group that appears with the name 4.0.B01**.

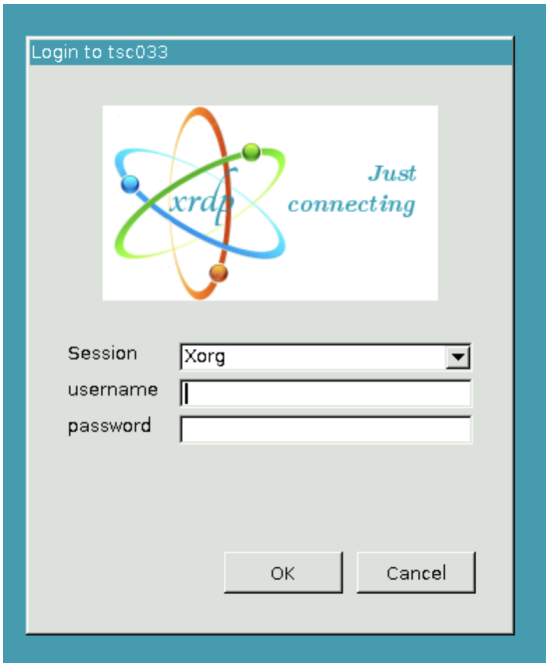
- Virtual resources, that is, virtual machines with Windows that will allow access to programs that require a license. **They are the ones that appear on the previous screen with the name Virtual Lab.**



In the case at hand, the option we must choose is the first, and we click on GPUs.

3.2 Access to equipment

As soon as you click on GPUs in the previous point, it will send us to the first free machine it finds in the group and will allow us to identify ourselves on it.



The data to access the selected machine will be those provided by the subject teacher. That is, we have three credentials in the system, we are going to review them:

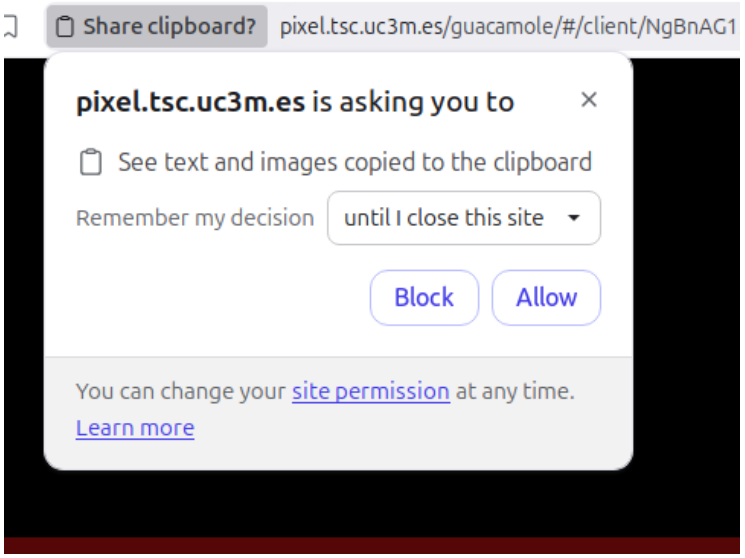
1. The university account, which we use to register in the system and generate a user for the remote laboratories.
2. The remote laboratories account from the previous point, which allows us to enter the system that coordinates remote resources.
3. The account provided by the subject teacher. What will be specific for that subject and will have assigned schedules.

In this last step, we must use the account of the third bullet.

3.3 Using remote equipment

When we enter the system, we are presented with the typical Linux-based desktop, with some considerations that we must take into account.

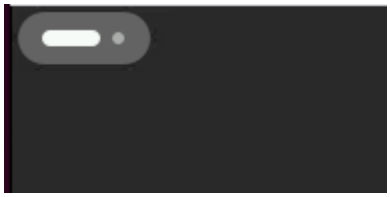
Also, depending on the browser used, it may ask us for access to the system clipboard to share information.

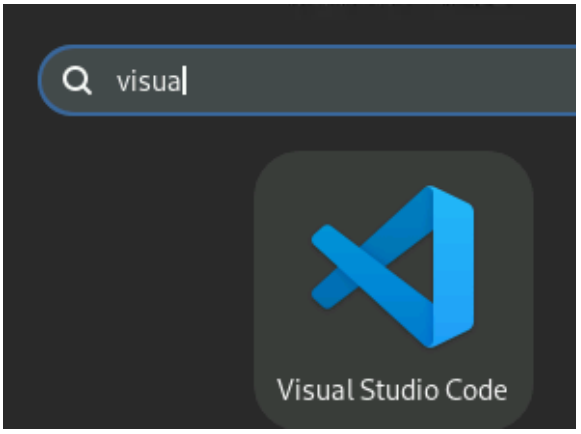


Remember that in order to use the clipboard the standard keys that we must use are:

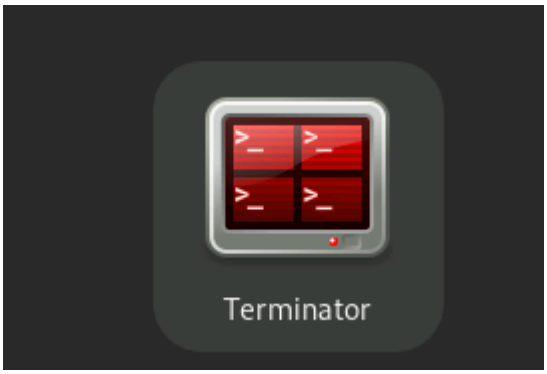
- On our desktop we will copy normally, using Control + C
- On the remote desktop, to paste the content, we must use SHIFT+Control+V

To access the programs installed in the system, we must click on the “activities” -which is represented as ‘-.’- button in the upper left corner, where we can write the program we want. For example, to access Visual Studio, we will click on “activities” and then start to write “visual”.

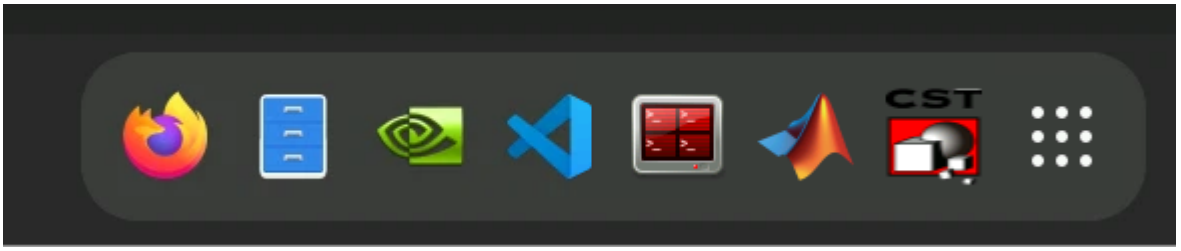




For typing commands in the terminal, we strongly recommend the use of terminator, instead of the standard terminal:



Remember that all the most common programs can be accessed directly through the taskbar at the bottom:



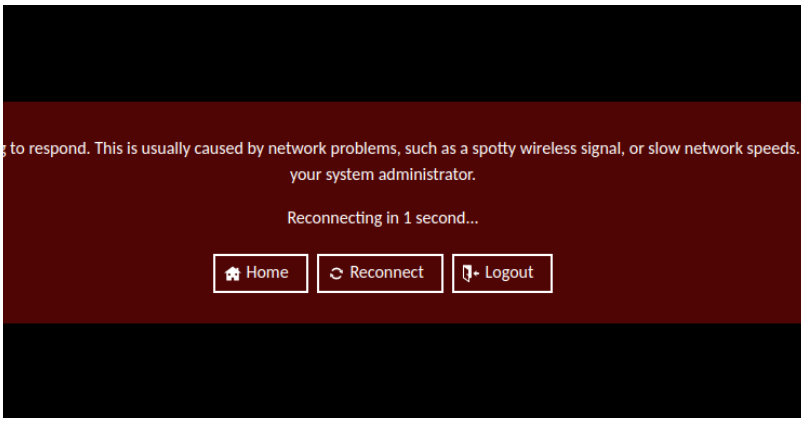
We also have the possibility of using Jupyter Notebook to edit code directly. To do this we must open a terminal and execute the command: “jupyter notebook”, which will automatically launch the system's default browser.

3.4 Saving data

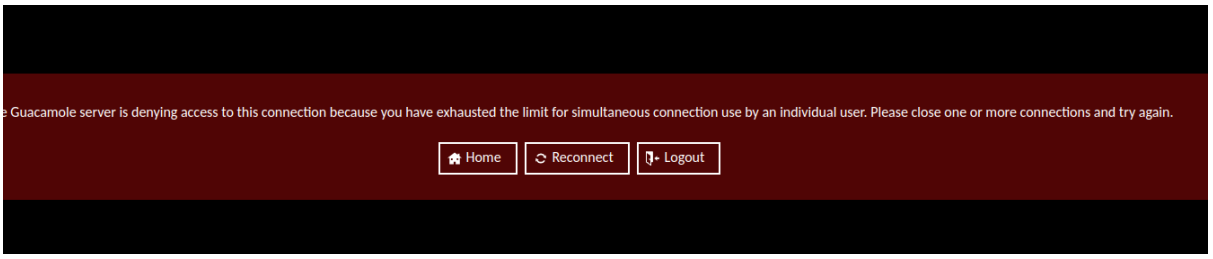
It is very important not to forget that we are in a shared environment on which maintenance is carried out, so **User data will not be saved from one session to the next**. **Don't forget to save all important information from your session**, for this you can use any file sharing system and, more reliably, use a Github repository.

3.5 Possible errors

If there is a problem with the network, a timeout error may occur in the system, so a message will appear allowing you to return to the virtual system or retry the connection, something that will happen automatically after ten seconds. You can wait that time or click on reconnect.



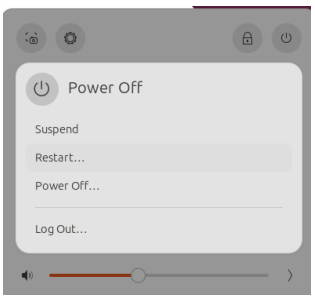
Remember that you cannot have more than one session open in the system. If you have opened more than one session with the virtual system (or have forgotten one), the system will remind you that you cannot have more than one session open:



If an error occurs during the session and you disconnect from it, the system will do its best to restart the session on the same computer with the same programs and data open.

3.6 Log out

It is advisable to exit the session using the exit provided by the operating system instead of directly closing the browser tab.

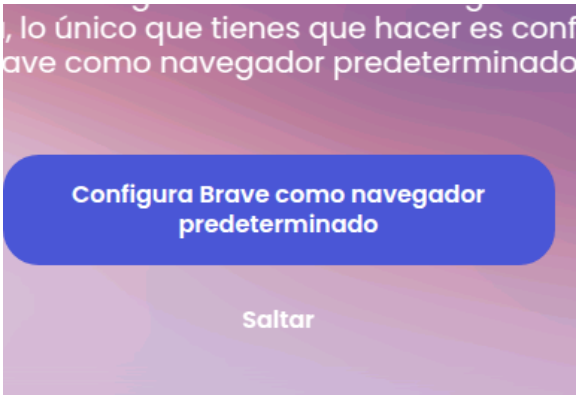

Power Off
Suspend
Restart...
Power Off...
Log Out...

4. Recommended Applications

Although it is generally possible to use all typical Unix system applications without issues, some applications perform better than others for remote access. Therefore, we recommend the following applications – all of which have direct access via the taskbar

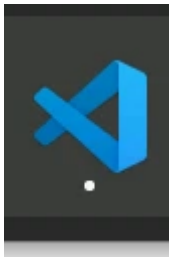
4.1 Web Browsing

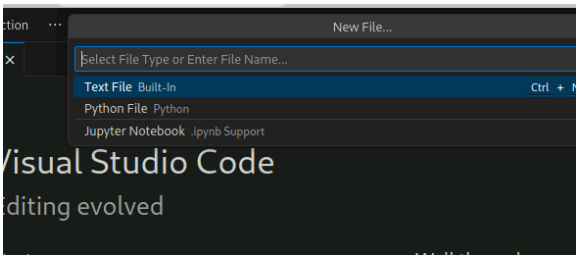
For opening web pages, we recommend using the Brave browser. Upon launch, it asks for some initial configuration data. Although the session is private for each user, it is advisable to log out of any open sessions once the work session is finished.



4.2 Programming Environments

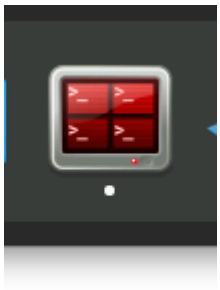
For editing programs in the Python environment, we recommend using the Visual Studio Code (VSCode) editor, which is pre-configured for editing and debugging in this programming language.





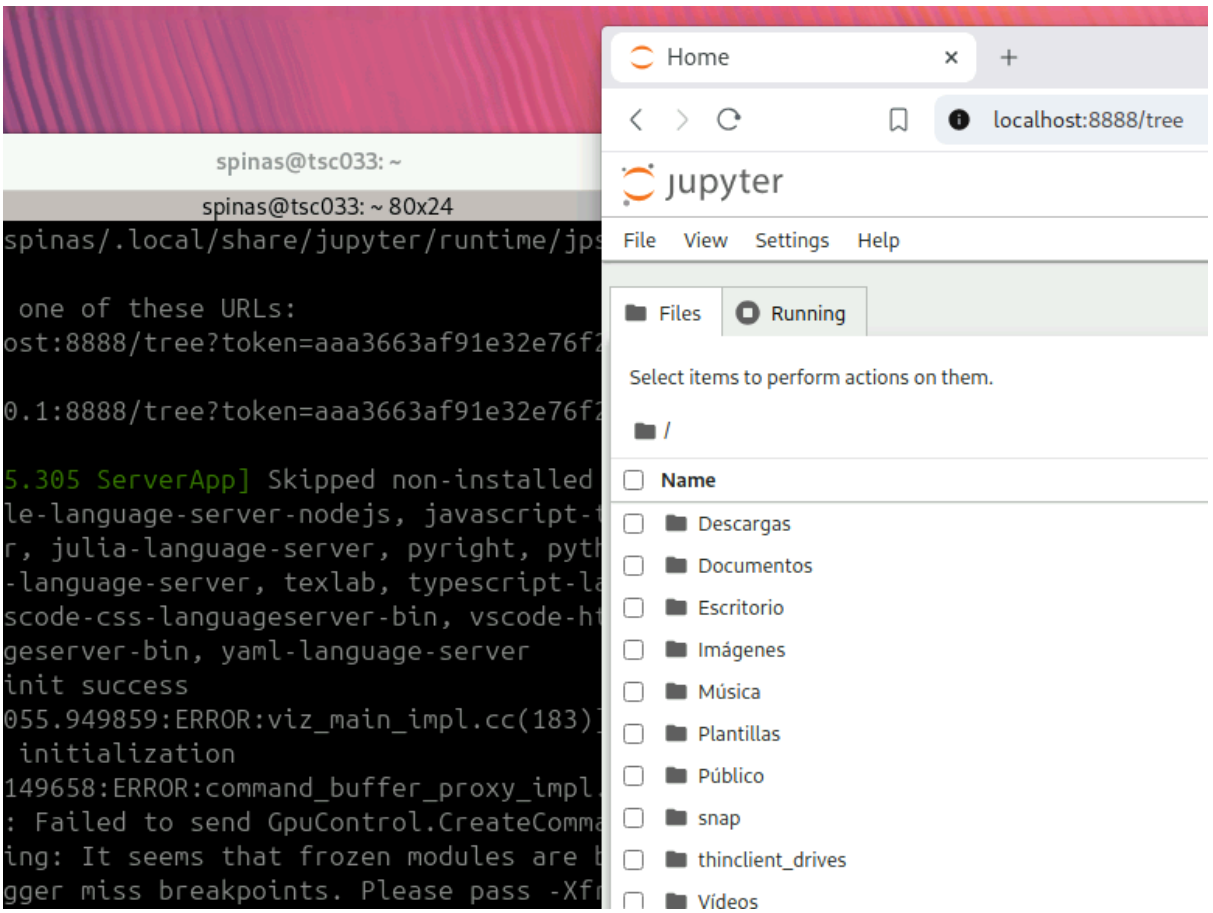
4.2 Terminal

If we need to run a command from the command line, we recommend using Terminator. It is a standard terminal, but for reasons beyond the scope of this document, it loads faster than the default program.



4.3 Python notebook:

The system allows launching Python notebooks. To do this, we must open a terminal (using Terminator) and execute the following command: jupyter notebook. Once the command is entered, and the environment is loaded, the system's browser will open with the notebook loaded.



Please note that any data saved during the session will be deleted within 24 hours. Any important information should be saved to your local machine.